Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/28/2006
PI Name:	Kasevich, Mark A. Ph.D.		
Project Title:	Atom Interferometry in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stanford	**State:**	CA
Zip Code:	94305-4060	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/1998	**End Date:**	01/01/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		

Light-pulse atom interferometry techniques are used to coherently manipulate atomic wavepackets. Sequences of light-pulses are used to create de Broglie wave interferometers which are capable of sensing inertial forces with high precision and accuracy. For example, high sensitivity rotation measurements are achieved in a geometry where a thermal Cs atomic beam propagates through three sets of spatially separated laser beams whose propagation axes are oriented perpendicular to the direction of the atomic velocity. High sensitivity measurements of gravitational accelerations are obtained by launching ensembles of atoms on vertical ballistic trajectories using laser cooling methods. Vertically propagating laser beams are pulsed on three times in order to measure ensemble accelerations. Gravitational gradient measurements are made by simultaneously launching two independent, vertically separated, ensembles on vertical ballistic trajectories. Travelling wave laser beams, whose propagation vectors pass through both ensembles, simultaneously record the acceleration of each ensemble, as well as their differential acceleration. The gradient signal is

Task Description:	obtained by dividing by the separation distance between the ensembles. Development of sensitive and accurate gravity gradiometers and gyroscopes using atom interference techniques. Precise and accurate inertial force sensing instruments are needed for navigation systems. The accelerometers, gyroscopes, and gravity gradient sensing instruments developed in this program could have significant impact on next-generation (2004 and beyond) navigation systems by improving both sensitivity and accuracy of key sensors by two orders of magnitude. In addition to outperforming current state-of-the-art sensors, our approach is likely to reduce sensor cost and improve reliability. Precise and accurate gravity gradiometers will find application in the oil/mineral industry (to aid in discovery and management of reservoir/deposits). Defense-oriented applications include underground structure detection, covert navigation, and gravity field mapping. Fundamental physics applications include improved tests of General Relativity and measurements of G. Exceptional instrument performance is expected in a microgravity environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)